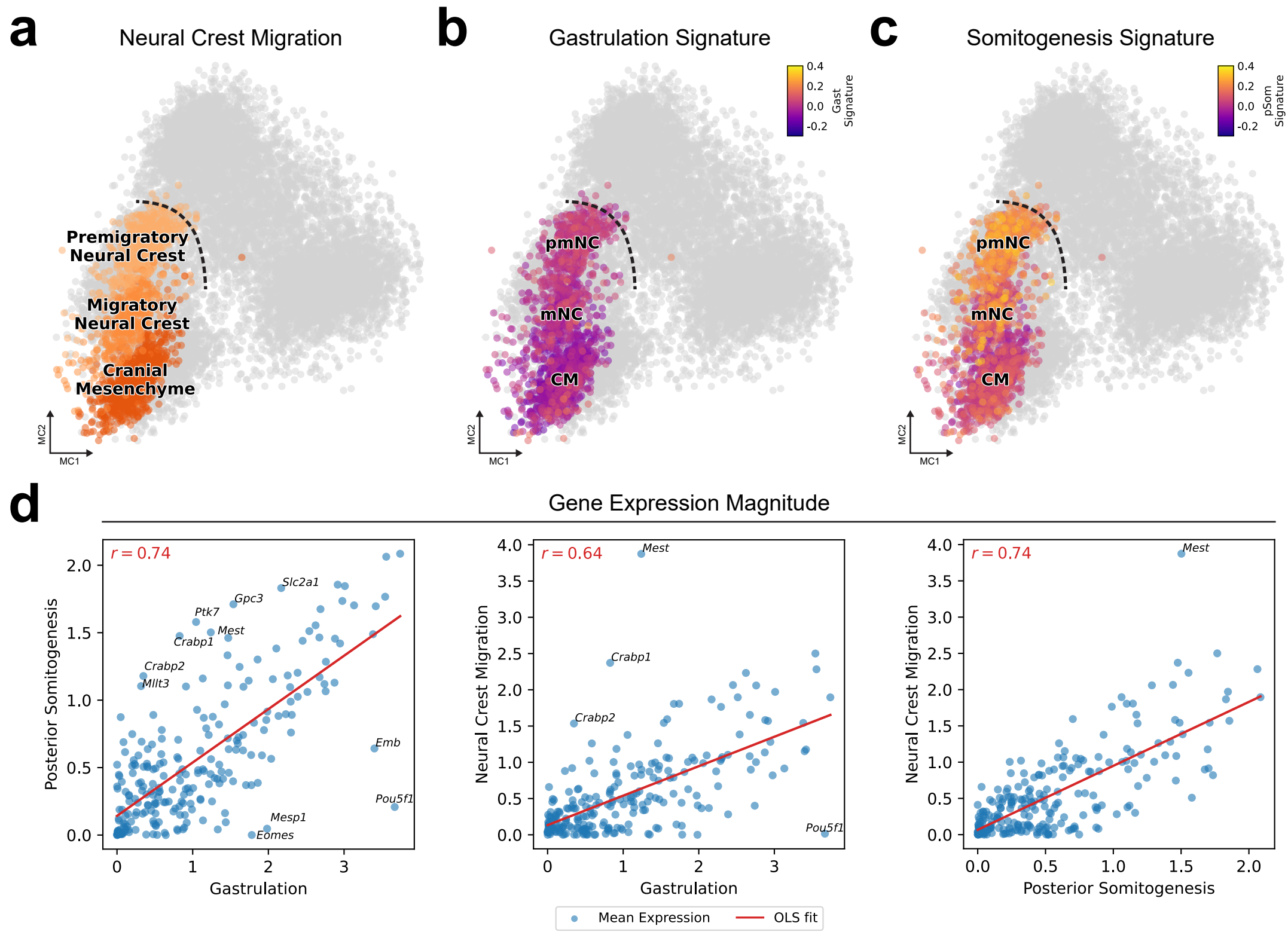

a
Neural Crest Migration
Premigratory Neural Crest
Migratory Neural Crest
Cranial Mesenchyme
MC2
MC1
b
Gastrulation Signature
0.4
0.2
0.0
-0.2
Gast Signature
pmNC
mNC
CM
MC2
MC1
c
Somitogenesis Signature
0.4
0.2
0.0
-0.2
pSom Signature
pmNC
mNC
CM
MC2
MC1
d
Gene Expression Magnitude
r = 0.74
Posterior Somitogenesis
Gastrulation
Slc2a1
Gpc3
Ptk7
Mest
Crabp1
Crabp2
Mllt3
Emb
Pou5f1
Mesp1
Eomes
r = 0.64
Neural Crest Migration
Gastrulation
Mest
Crabp1
Crabp2
Pou5f1
r = 0.74
Neural Crest Migration
Posterior Somitogenesis
Mest
Mean Expression
OLS fit
e
Gene Expression Timing
Posterior Somitogenesis
Late
Early
Early
Late
Gastrulation
Meox1
Tcf15
Ebf3
Foxc2
Tmem200b
Six2
Arg1
Six1
Aldh1a2
Ntn1
Itga9
Twist1
Sertad4
Efnb2
Msx1
Tril
Nin
Lix1
Nkx1-2
Hes3
r = 0.54
Neural Crest Migration
Late
Early
Early
Late
Gastrulation
Sertad4
Ppfibp2
Crym
Sh2d3c
Bmpr1b
Aldh1a2
Efna3
T
Lfng
Wnt3a
Sp5
Ntn1
Hes3
Dll1
Foxb1
r = 0.49
Neural Crest Migration
Late
Early
Early
Late
Posterior Somitogenesis
Hand1
Crym
Fam43a
Ppfibp2
Tril
Mesp2
Aldh1a2
Lfng
Nanog
Dll1
Ntn1
Foxb1
r = 0.48
Mean PseudoTime
OLS fit
f
Pou5f1
Eomes
Mesp1
Mest
Crabp1
Crabp2
Rarb
Epi
pEpi
LPM
NE
NMP
PSM
Somite
pmNC
mNC
NGP
CM
% Cells
10 20 30 40 ≥50
Gene Expression Score
0 1 2 3
g
Somitogenesis
Meis2
Meis1
Crabp1
Rarg
Fabp5
Aldh1a2
Crabp2
Rbp1
h
Neural Crest
Rarg
Rarb
Rbp1
Meis2